

**VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2008
Higher 2**

CANDIDATE NAME

CLASS

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INDEX NUMBER

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BIOLOGY

9747/03

Paper 3

17 September 2008

Additional Materials: Answer Booklet/ Paper

1 Hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your CT GP/ INDEX NO. and name on all the work you hand in.
Write in dark blue or blue pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use any staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The intended number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Total	

This paper consists of **12** printed pages.

Answer **all** questions.

- 1 In humans, *EcoRI* site polymorphisms distinguish wild type (D) from mutant (d_i , d_{ii} , d_{iii}) alleles of a gene associated with an autosomal recessive disease:

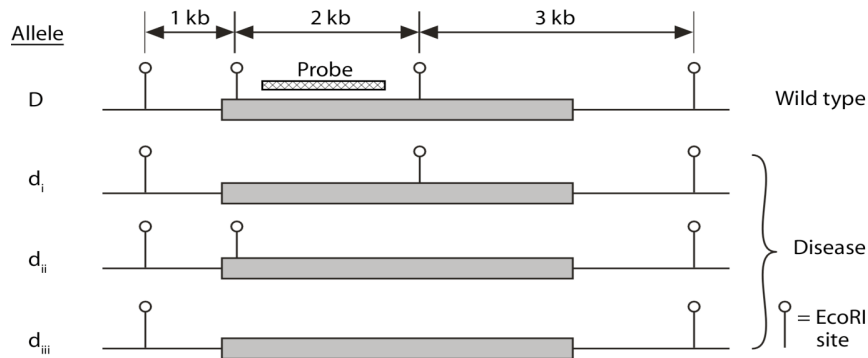


Fig.1.1

- (a) A researcher wants to investigate the prevalence of the disease in his fiancé's family. The pedigree from individuals in her extended family in which some family members show the disease is shown below. **Figure 1.2** shows the results of testing for the disease gene. Each person's DNA sample is shown directly below that person.

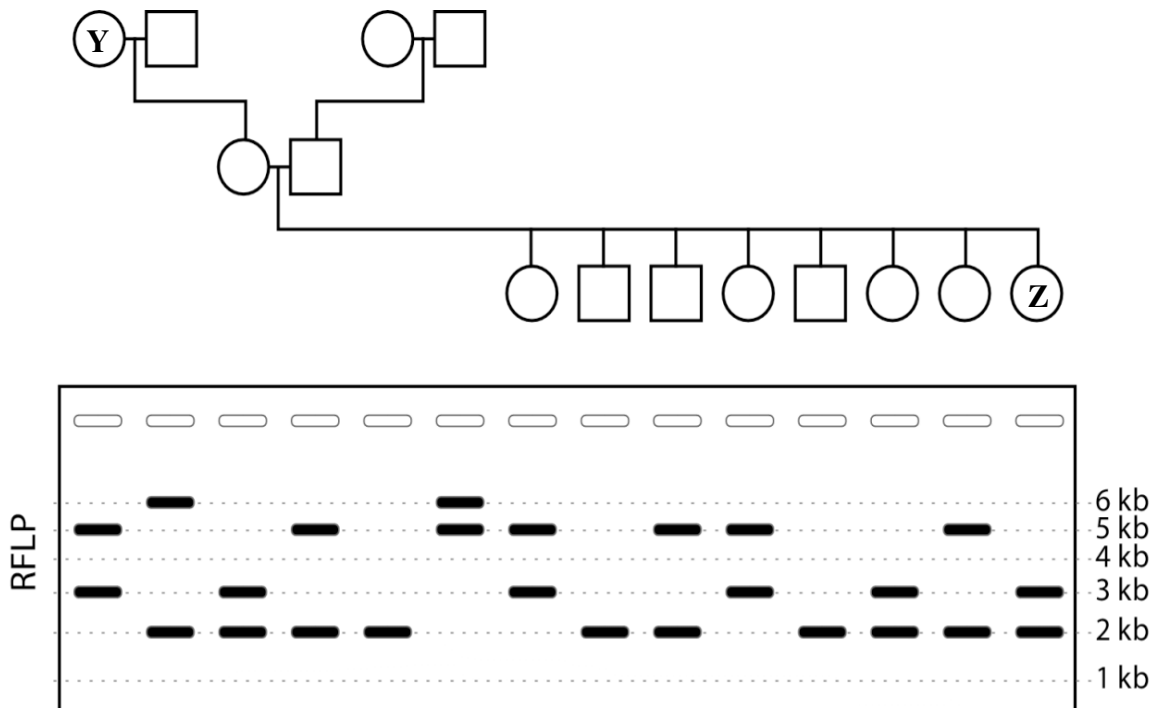


Fig. 1.2

(i) Describe how the bands in **Figure 1.2** were obtained.

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..... [3]

(b)(i) Which individual in the F2 generation is likely to be a carrier for the disease? Mark out ONE individual with an **X** in the pedigree diagram in **Figure 1.2**. [1]

(ii) If individual **Y** and spouse were to have another child, what is the probability that the child would be a male with the disease?

..... [1]

(iii) With reference to **Figure 1.2**, explain, with reasons, how one could diagnose if individual **Z** will come down with the disease.

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- (c) Prior to the discovery of the identity of the gene responsible for this disease, prediction of the disease used to be based on analysis of whether the individual inherited specific alleles from a separate locus which displays microsatellite repeat polymorphism.

Explain what is meant by microsatellite repeat polymorphism and how it was used to facilitate the process of disease detection.

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.....[3]

- (d) Based on similar principles used for the detection of the disease in his fiancée's family, the researcher develops a method to detect for the presence of the *ras* oncogene which is involved in a variety of human and animal cancers. DNA was isolated from a number of normal and cancerous tissues and with *EcoRI*.

Fig. 1.3 shows the autoradiograph result for testing for the *ras* oncogene

- 1) white blood cells from a healthy human
- 2) human lymphoma cells (cancerous white blood cells)
- 3) human bladder carcinoma cells (cancerous)
- 4) human lung sarcoma cells (cancerous)
- 5) white blood cells from a healthy mouse
- 6) mouse lymphoma cells (cancerous white blood cells)

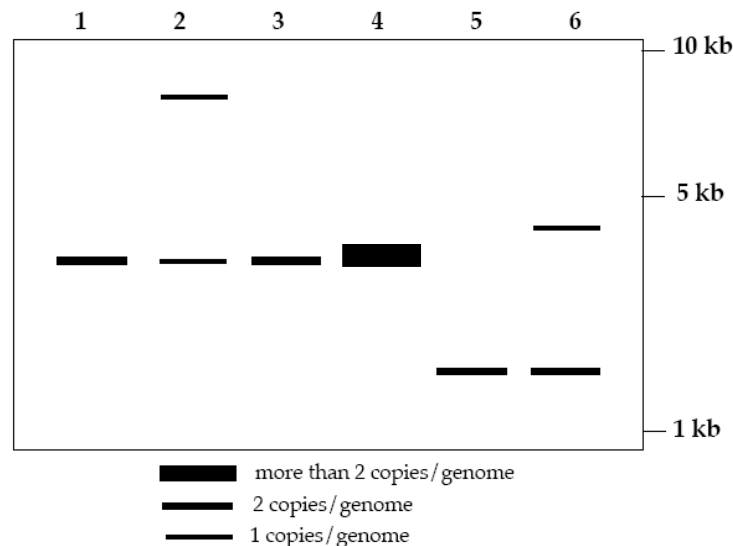


Fig. 1.3

- (i) The healthy human (1) and mouse (5) controls used in the test do **not** have cancer. Explain why when their DNA is analysed, the autoradiogram result for the test for the ras oncogene yields positive band results just like that for the cancerous samples.

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..... [2]

- (ii) With reference to **Fig. 1.3** briefly explain how the process of the transformation of cell **3** to cancerous bladder carcinoma cells differs from the transformation of cell **4** to form lung cancer cells.

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- (e) Explain briefly why the products of digestion of DNA from the same animal is different when a different restriction enzyme is used. [2]

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..... [2]

[Total: 16]

- 2** Severe combined immunodeficiency (SCID) is caused by a severe genetic defect often found in newborns. The condition must be diagnosed and treated quickly to prevent serious complications, and doctors continue to struggle with often ineffective treatment options.

In some studies, researchers found that blood stem cells, better known as hematopoietic stem cells (HSCs), may effectively fight SCID caused by a deficiency of the ADA gene (adenosine deaminase), which is critical for the immune system to function properly.

- (a) (i)** Explain how a person with SCID caused by ADA deficiency inherited this condition.

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 [2]

- (b) (i)** Describe how the normal functions of adult stem cells obtained from the bone marrow and stem cells obtained from a zygote differ.

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 [2]

- (ii)** Explain the how the properties of hematopoietic stem cells allow them to be effectively used to fight a disease such as SCID.

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 [3]

A trial was carried out on a ADA-SCID patient of age 7 months. Hematopoietic stem cells (HSCs) were collected from the patient's bone marrow, transduced with retroviral vector containing the ADA gene, and infused back. The patient received 8.6×10^6 HSCs per kg body weight, containing 25% successfully transformed cells in culture.

Figure 2.1A below shows the number of total lymphocyte counts (solid circles) (left ordinate) and white blood cell counts (open circles) (right ordinate) in the peripheral blood (PB) for this patient.

DNA was also isolated and analyzed for proportion of transformed cells. **Figure 2.1B** shows the percent of transformed cells in the following sorted cell populations: T cells (open circles) and NK cells (solid circles).

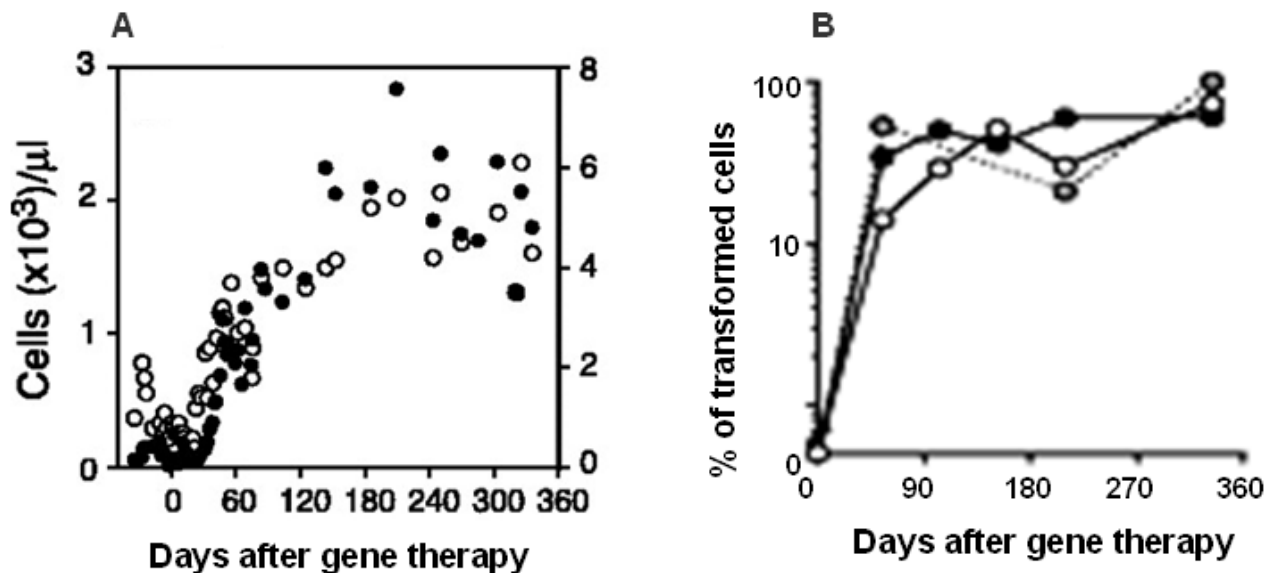


Figure 2.1

- (c) (i) What is a medical advantage of using the patient's own hematopoietic stem cells rather than stem cells from a donor?

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..... [1]

(ii) Suggest why a retroviral vector was chosen for use in this trial.

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(d) Using evidence from **Figure 2.1**, explain why haematopoietic stem cells in particular are used for gene therapy.

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[Total: 14]

- 3 Plant tissue culture or micropropagation is a method used to propagate plants. The diagram below shows one method of plant tissue culture.

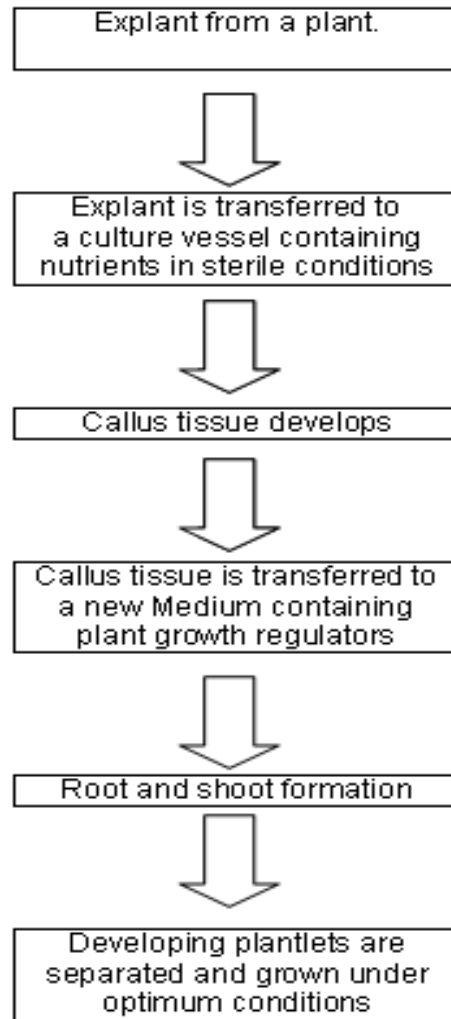


Fig. 3.1

- (a) Explain what is meant by an explant?

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[1]

- (b) Explain how plant cells may be grown into a callus tissue.

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[2]

- (c) Explain why a viral infection is more likely to destroy a complete batch of plants grown by plant tissue culture than a batch of plants grown from seeds.

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.....[3]

Dinitroaniline herbicides act by binding to the protein tubulin, thereby disrupting the formation of spindle fibres. Grasses resistant to dinitroaniline seemed to carry a substitution mutation in the gene **Tub-1** (the gene for tubulin) such that, at one site in the tubulin protein, the amino acid threonine was replaced by isoleucine.

An experiment was performed on cells of oil seed rape (*Brassica napus*) to see whether such a mutation in grass was indeed responsible for their resistance toward dinitroaniline. Oil seed rape cells were grown in tissue culture and divided into four groups (**M2**, **M1**, **N** and **C**) which were treated as follows:

Groups	Treatment
M2	Genetically engineered to express two copies of the mutated Tub-1 gene from grass.
M1	Genetically engineered to express one copy of the mutated Tub-1 gene from grass.
N	Genetically engineered to express the normal Tub-1 gene from grass.
C	Was not genetically engineered.

Table 3.2

- (d) Besides viruses and gene gun, briefly describe another method how the **Tub-1** gene from grass was introduced into oil seed rape cells.

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.....[3]

Each group of cells was then grown into callus and equal quantities of each callus were then grown in the presence of different concentrations of dinitroaniline herbicide. The table below shows the results of the experiment. The shaded circles show the relative size of each piece of callus tissue after growth in the presence of dinitroaniline.

maize callus	dinitroaniline herbicide concentration (mg dm^{-3})				
	0.00	0.05	0.10	0.50	1.00
M2					
M1					
N					
C					

Table 3.3

- (e) Explain why oil seed rape calluses N and C were used in this investigation?

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.....[2]

- (f) Account for the results shown in **Table 3.3** and conclude if the mutated **Tub-1** gene from grass is responsible for dinitroaniline resistance.

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.....[4]

[Total: 15]

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Write your answers to this question on the separate answer paper provided. Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be set out in sections (a), (b) etc. as indicated in the question.

- 4 (a)** Outline the procedure for cloning the human growth factor gene using *E.coli* cells to produce the human growth factor protein. Your answer should include how the eukaryotic gene is cloned to avoid the problems associated with introns. [12]

(b) Compare the similarities and differences in the function and properties of bacterial plasmid and adenovirus. [4]

(c) Discuss the benefits and ethical concerns of the human genome project.[4]

[Total: 20]